DP Mesh Dashboard

QUICK SETUP GUIDE · LOCAL INSTALLATION & VIEWING

Purpose: Install and open the dashboard locally on another computer. This guide covers only the software requirements and running the website; it does not cover ML/API integration.

1. Requirements

Required: Node.js LTS (Node.js 20 LTS is a suitable choice; npm is included) and a modern browser such as Chrome, Edge, Firefox, or Safari. Git is optional if you share the project through a repository. Python is not required to run the dashboard.

Project packages: React, React DOM, Leaflet, Leaflet Heat, Vite, the Vite React plugin, and the Vite PWA plugin. These are installed automatically by npm; there is no need to install them globally.

2. Install and run the dashboard

Step 1 — Extract the ZIP. Extract `DP_Mesh_Dashboard_Handoff.zip` to a folder on your computer.

Step 2 — Open a terminal. Navigate into the `mesh-dashboard` directory, the folder containing `package.json`. On Windows, use PowerShell or Windows Terminal; on macOS, use Terminal.

Step 3 — Install dependencies.

```
npm install
```

Step 4 — Start the development server.

```
npm run dev
```

Step 5 — View the website. Open the URL printed in the terminal in a browser. It is usually `http://localhost:5173`. Keep the terminal running while viewing the dashboard. Press `Ctrl+C` in the terminal to stop it.

3. What to expect

The dashboard can display its demo interface using mock node and telemetry data. These are sample values, not live readings from physical devices. The map's OpenStreetMap tiles require internet access; other parts of the interface may still load if map tiles are unavailable.

4. Optional production preview

To create a production build, run:

```
npm run build
```

To preview the generated build locally, run:

```
npm run preview
```

Open the local URL printed by the preview command. This is a local preview, not a permanently hosted public website.

5. Troubleshooting

- `npm` **not found:** Install Node.js LTS, reopen the terminal, and check `node -v` and `npm -v`.
- **Website does not open:** Make sure `npm run dev` is still running and use the exact URL shown in the terminal.
- **Dependency errors:** Confirm you are inside the folder containing `package.json`, then rerun `npm install`.
- **Map tiles missing:** Check the internet connection.

Project documentation

For additional details, see `mesh-dashboard/README.md` inside the extracted project folder.